B - General MS4092

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28 Mei 2021
Number of problems: 8
Start at: 14:00 PM
Finish at: 14:21 PM
Duration: 21 Minutes
If you think there is no correct answer, please choose the closest answer.

if -x^2y-6xy+y^2=0, what is the value of x that yields the maximum/minimum value of y?

- -3
- -6
- 3
- 6

$-x^2y' - 2xy - 6xy' - 6y + 2yy' = 0$

$2xy + 6y = (2y - 6x - x^2)y'$

$y' = \frac{2xy+6y}{2y-6x-x^2}$ Max/min → $\frac{dy}{dx} = 0$

$2xy + 6y = 0$

$(2x+6)y = 0$

$x = -3$

In technical drawing, a short dash lines used to represent .....

- outline of objects, visible edges and surfaces.
- a cutaway view of a section
- a hidden or non-visible surface
- none of the other answers are correct

Which from these following parameters of the sensor that is more important than the others, if you intend to measure the change of temperature of a brass-cylinder that rapidly cools down from 100 degree Celsius to a room temperature?

- Thermocouple response time and accuracy.
- Thermocouple accuracy and hysteresis.
- Thermocouple response time and hysteresis.
- Thermocouple accuracy and resolution.

Evaluate the following limit

$$\lim_{x\to 4}\left(\frac{x^2-16}{x-4}\right)$$

$= \lim_{x\to 4} \frac{(x+4)(x-4)}{x-4}$

$= 8$

- 0
- 4
- ~~8~~
- infinity

Consider a relation between temperature and time which is correlate by dT/dt = 5t – 10 where T is in Celsius and t is in second. The temperature will reach its steady state at?

Steady state : $\frac{dT}{dt} = 0$

$5t - 10 = 0$

$t = 2\,s$

- ~~2 seconds~~
- 5 seconds
- 10 seconds
- Never reach steady state

An area in 2D plane is bounded by y = 0; y = -x^2+8x+16; x = 2; and x = 4. If two-segments trapezoidal rule is used to estimate this area, find the difference between the estimated and the correct value of the area!

$\int_2^4 -x^2+8x+16 = \left[-\frac{x^3}{3} + 4x^2 + 16x\right]_2^4$

$= \frac{320}{3} - \frac{136}{3} = \frac{184}{3}$

$\Delta x = \frac{4-2}{2} = 1$

$x_i = a + i.\Delta x$

$x_0 = 2$

$x_1 = 3$

$x_2 = 4$

$f(x_0) = 28$

$f(x_1) = 31$

$f(x_2) = 32$

$T_n = \frac{\Delta x}{2}\left[f(x_0) + 2f(x_1) + f(x_2)\right]$

$= \frac{1}{2}\left[28 + 62 + 32\right] = 61$

Perbedaan $= \frac{184}{3} - 61$

$= \frac{1}{3}$

- 2/3
- ~~1/3~~
- 1/2
- 1

A program shown below. If the a = 3 and b = 1, the value of m will be?

```
x = a
n = 0
m = 0
        do while n < x
        m = m*x*n+x^n
        n = n + b
        end
print m
```

n=0
m=1
n=1
m=6
n=2
m=45

- 432
- 45
- 6
- 1

Consider a three-phase star AC motor with line-line voltage of V, line-line current of I, and power factor of PF. Which of the following statement is correct?

- The power (in kVa) of the motor is calculated by VI cos PF.
- The power (in watt) of the motor is calculated by VI cos PF.
- The power (in kVa) of the motor is calculated by 1.73 VI cos PF.
- None of the other answers are correct.

Kembali Kirim

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